

Recommendation and Review Analysis Report

Market Basket Analysis with Python– Amazon

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Data Set	Amazon Customer Behaviour Survey
Language	Python
Libraries	Numpy, Pandas, Matplotlib, Seaborn, scipy, sklearn

Project Overview

This report presents a comprehensive analysis of amazon customer behaviour analysis using the Amazon Customer Survey Analysis Survey. The analysis covers data cleaning, descriptive behaviour analysis, customer segmentation and profiling, recommendation and review insights and visualization.

Business Problem Addressed

- What products or categories are frequently purchased together, indicating cross-selling opportunities?
- How do customer purchasing patterns vary across demographics or regions?
- Can we segment customers based on buying behaviour to target personalized promotions?
- Which products drive repeat purchases or high customer loyalty?

Data Cleaning and Preparation:

- Handled Null Values
- Fixed Duplicated columns by renaming
- Cleaned column names

Cleaned Data:

RangeIndex: 800 entries, 0 to 799

Data columns (total 28 columns):

#	Column	Non-Null Count	Dtype	Timestamp	0
0	Timestamp	800 non-null	object	age	0
1	age	800 non-null	int64	Gender	0
2	Gender	800 non-null	object	Purchase_Frequency	0
3	Purchase_Frequency	800 non-null	object	Purchase_Categories	0
4	Purchase_Categories	800 non-null	object	Recommendation_Response	0
5	Recommendation_Response	800 non-null	object	Browsing_Frequency	0
6	Browsing_Frequency	800 non-null	object	Product_Search_Method	0
7	Product_Search_Method	800 non-null	object	Search_Result_Exploration	0
8	Search_Result_Exploration	800 non-null	object	Customer_Reviews_Importance	0
9	Customer_Reviews_Importance	800 non-null	int64	Add_to_Cart_Browsing	0
10	Add_to_Cart_Browsing	800 non-null	object	Cart_Completion_Frequency	0
11	Cart_Completion_Frequency	800 non-null	object	Cart_Abandonment_Factors	0
12	Cart_Abandonment_Factors	800 non-null	object	Saveforlater_Frequency	0
13	Saveforlater_Frequency	800 non-null	object	Review_Left	0
14	Review_Left	800 non-null	object	Review_Reliability	0
15	Review_Reliability	800 non-null	object	Review_Helpfulness	0
16	Review_Helpfulness	800 non-null	object	Recommendation_Score	0
17	Recommendation_Score	800 non-null	int64	Recommendation_Helpfulness	0
18	Recommendation_Helpfulness	800 non-null	object	Rating_Accuracy	0
19	Rating_Accuracy	800 non-null	int64	Shopping_Satisfaction	0
20	Shopping_Satisfaction	800 non-null	int64	Service_Appreciation	0
21	Service_Appreciation	800 non-null	object	Improvement_Areas	0
22	Improvement_Areas	800 non-null	object	transaction	0
23	transaction	800 non-null	int64	Age_Group	0
24	Age_Group	800 non-null	category	Purchase_Frequency_Rank	0
25	Purchase_Frequency_Rank	800 non-null	int64	Customer_Segment	0
26	Customer_Segment	800 non-null	object	Cluster	0
27	Cluster	800 non-null	int32	dtype: int64	

Descriptive Behaviour Analysis :

Insights:

- Senior Citizen over 50+ age showing the most engagement.
- Also the working class of adult and young adult is also contributing significantly
- Clothing and Fashion is the most popular category among customers
- Mostly customers abandon cart due to high shipping costs
- The frequency of shopping customer is low among the customers

Recommendation:

- Amazon must keep clothing and fashion related product focused on senior and working class.
- Amazon need to first look upon their shipping cost as well as other factor to reduce cart abandonment reasons.

Customer Segmentation and Profiling

Insights:

- The data is almost uniform among the age group, shopping satisfaction and purchase frequency which means all are contributing equally for amazon
- Most customers are occasional customers

Recommendation:

- To increase customer engagement there is to increase sale, also highlighting products with less high shipping cost and more discounts.
- To increase customer activity amazon should add clothing and fashion related products and also introduce offers on the same

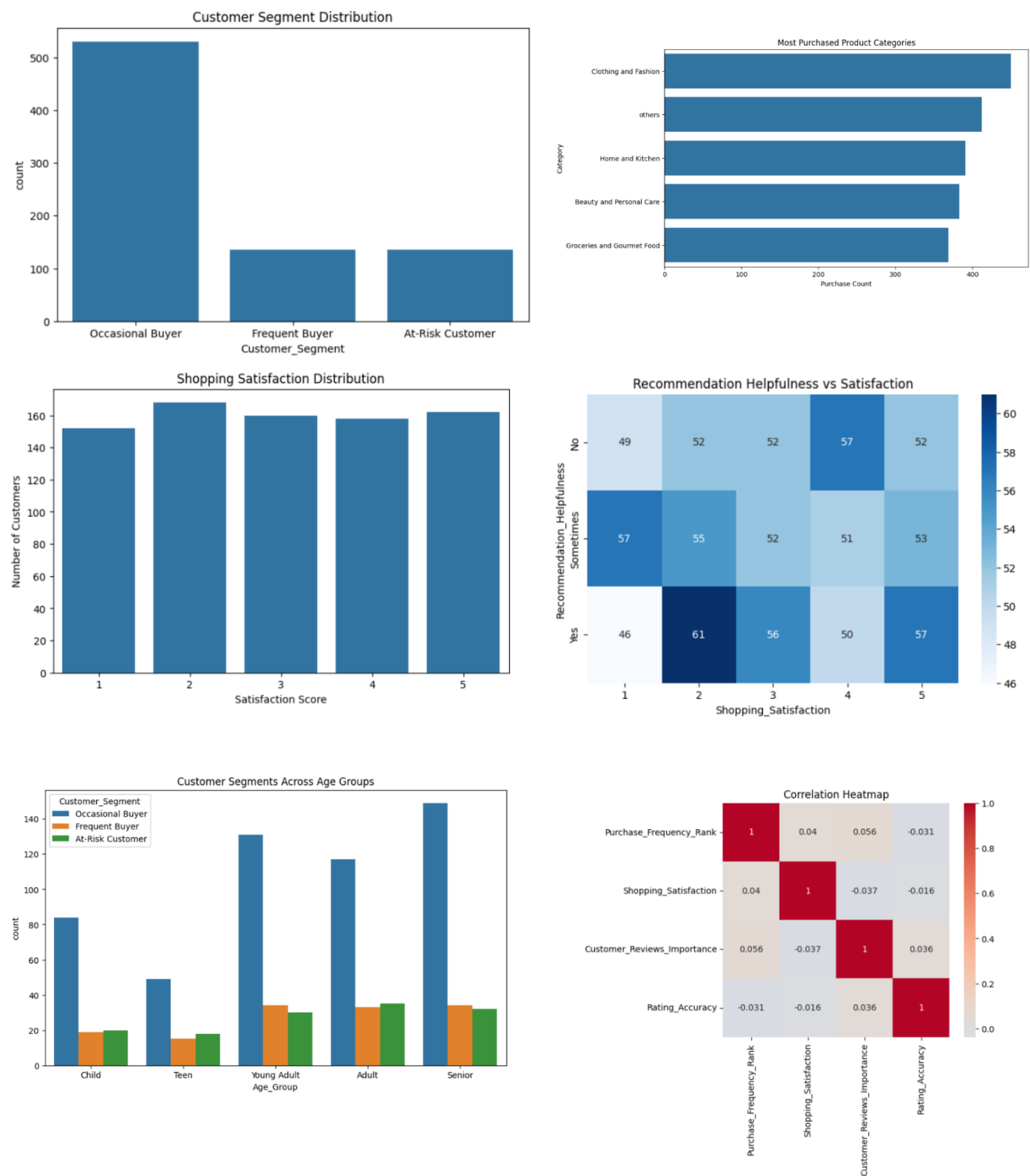
Recommendation and Review Insights

Insights:

- The recommendation is not showing any significance difference on shopping satisfaction
- The satisfaction average is high (3.20) for the customers moderately depend upon review while the satisfaction for customer which heavily depend upon review have less satisfaction(3.02)
- The reviews are increasing expectation which should be not delivered due to which shopping satisfaction is low for the customers heavily depend upon reviews.

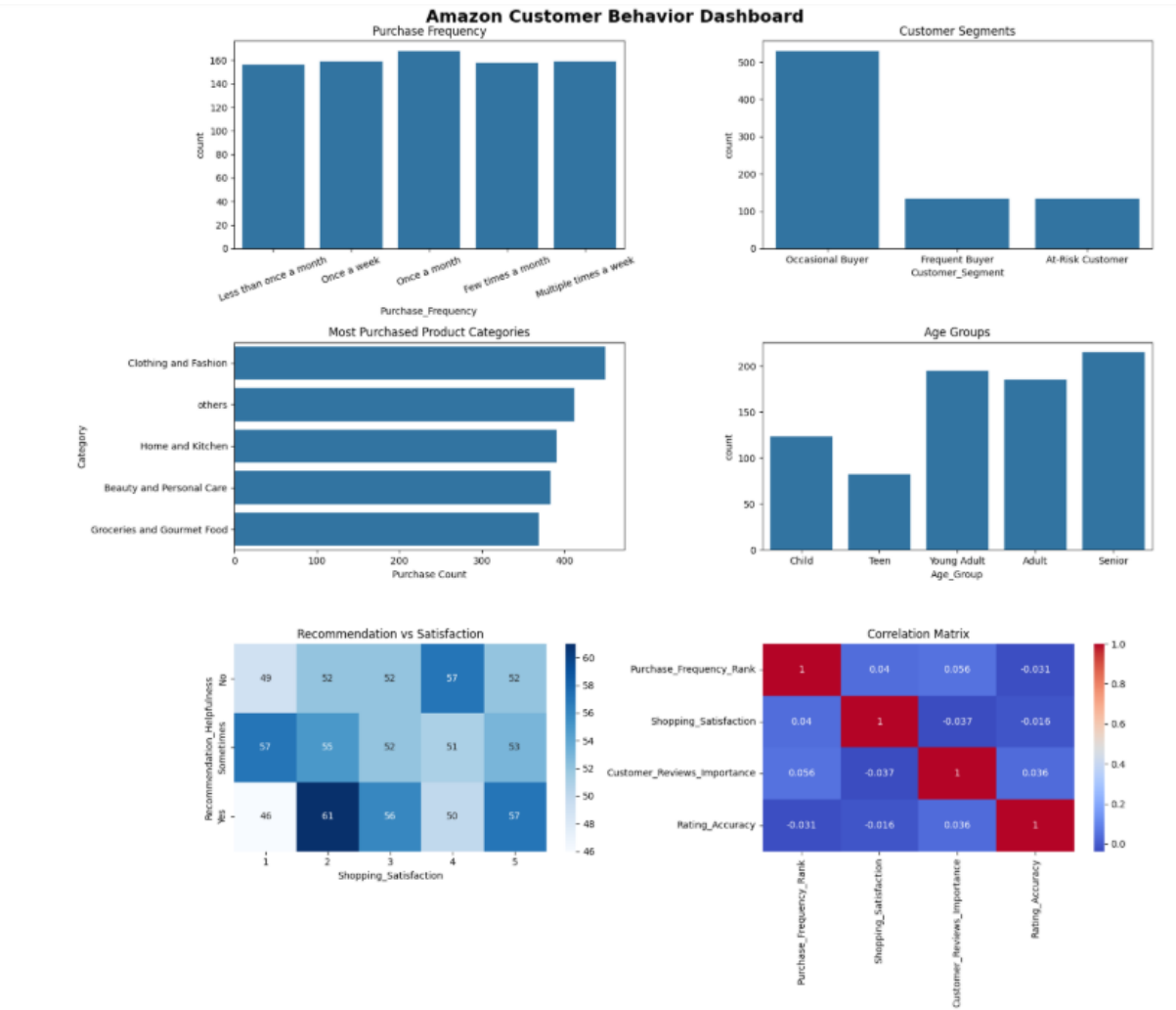
Visualisation and Dashboarding

Visualisation:



Dashboard:

Total Customers : 800
Average Satisfaction : 3.01
Frequent Buyers : 135
At-Risk Customers : 135



Task 8 : Video and project link

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